Supplementary information

Appellian Neural Networks

Serhii Bahdasariants, Lauren Parola, Kriti Kacker, Ariel K. Feldman, Zachary Zdobinski, Inseung Kang, Douglas J. Weber

Forward dynamics equations

Let $\mathbf{q} = [q_1, q_2]^T$ denote joint angles and $\dot{\mathbf{q}} = [\dot{q}_1, \dot{q}_2]^T$ denote angular velocities. From the Gibbs–Appell formulation, the partial derivative of the Appell acceleration energy with respect to generalized acceleration can be written as:

$$\frac{\partial S}{\partial \ddot{\mathbf{q}}} = M(\mathbf{q})\ddot{\mathbf{q}} + F_c(\mathbf{q}, \dot{\mathbf{q}}), \tag{S1}$$

where $M(\mathbf{q})$ is the configuration-dependent mass matrix and $F_c(\mathbf{q}, \dot{\mathbf{q}})$ contains velocity-dependent inertial terms. Setting this expression equal to the negative gradient of the potential energy gives:

$$M(\mathbf{q})\ddot{\mathbf{q}} + F_c(\mathbf{q}, \dot{\mathbf{q}}) = -G(\mathbf{q}), \tag{S2}$$

where $G(\mathbf{q}) = \partial V/\partial \mathbf{q}$. Thus, the ground-truth forward dynamics used for numerical integration were obtained as:

$$\ddot{\mathbf{q}} = -M(\mathbf{q})^{-1}[F_c(\mathbf{q}, \dot{\mathbf{q}}) + G(\mathbf{q})]. \tag{S3}$$

For the two-link system, the matrix and vectors were:

$$M(\mathbf{q}) = \begin{bmatrix} l_1^2(m_1 + m_2) + 2l_1 l_2 m_2 \cos q_2 + l_2^2 m_2 & l_2 m_2 (l_1 \cos q_2 + l_2) \\ l_2 m_2 (l_1 \cos q_2 + l_2) & l_2^2 m_2 \end{bmatrix}, \tag{S4}$$

$$F_c(\mathbf{q}, \dot{\mathbf{q}}) = \begin{bmatrix} -l_1 l_2 m_2 \sin q_2 \, (2\dot{q}_1 \dot{q}_2 + \dot{q}_2^2) \\ l_1 l_2 m_2 \sin q_2 \, \dot{q}_1^2 \end{bmatrix}, \tag{S5}$$

$$G(\mathbf{q}) = \begin{bmatrix} g[l_1(m_1 + m_2) \sin q_1 + l_2 m_2 \sin(q_1 + q_2)] \\ g l_2 m_2 \sin(q_1 + q_2) \end{bmatrix}. \tag{S6}$$

Equivalently, defining $D = m_{11}m_{22} - m_{12}^2 = l_1^2 l_2^2 m_2 (m_1 + m_2 \sin^2 q_2)$, the joint accelerations were:

$$\ddot{q}_1 = \frac{-m_{22}\left(F_{c_1} + G_1\right) + m_{12}\left(F_{c_2} + G_2\right)}{D}, \tag{S7}$$

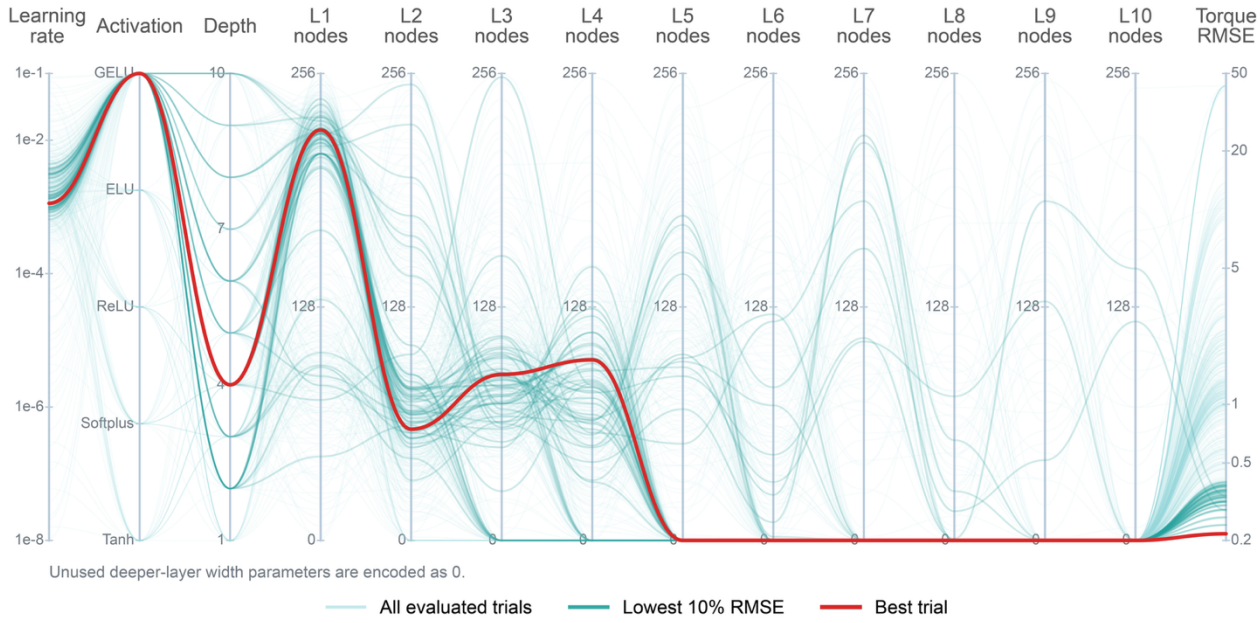
$$\ddot{q}_2 = \frac{m_{12}\left(F_{c_1} + G_1\right) - m_{11}\left(F_{c_2} + G_2\right)}{D}. \tag{S8}$$

These accelerations were then reformulated as the first-order system:

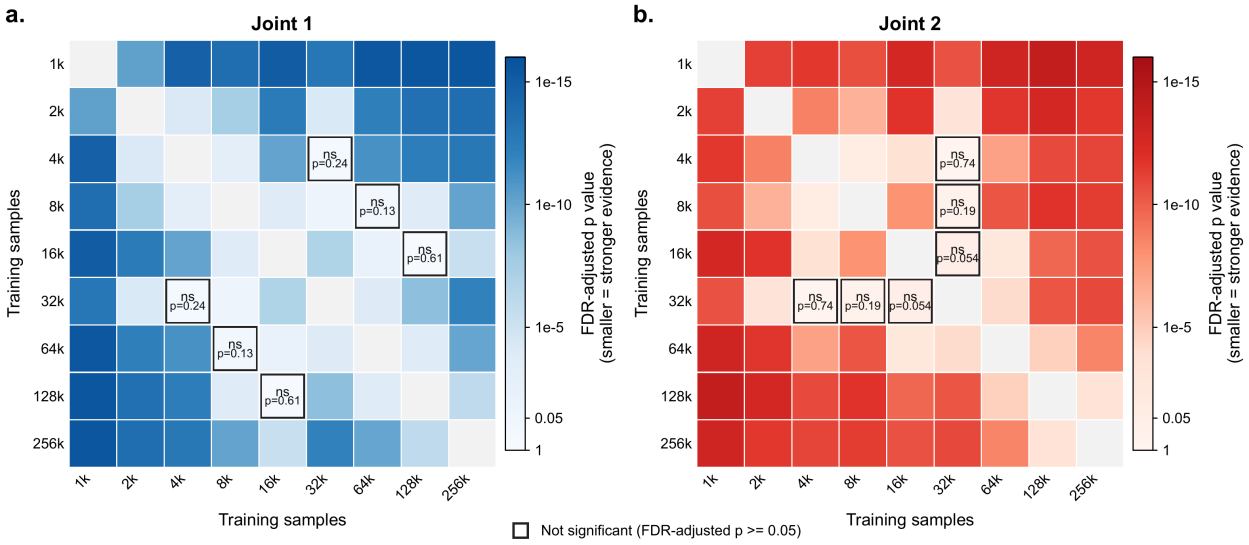
$$\frac{d}{dt}\begin{bmatrix} q_1 \\ q_2 \\ \dot{q}_1 \\ \dot{q}_2 \end{bmatrix} = \begin{bmatrix} \dot{q}_1 \\ \dot{q}_2 \\ \ddot{q}_1(\mathbf{q}, \dot{\mathbf{q}}) \\ \ddot{q}_2(\mathbf{q}, \dot{\mathbf{q}}) \end{bmatrix}, \tag{S9}$$

which was integrated using the fixed-step fourth-order Runge–Kutta scheme.

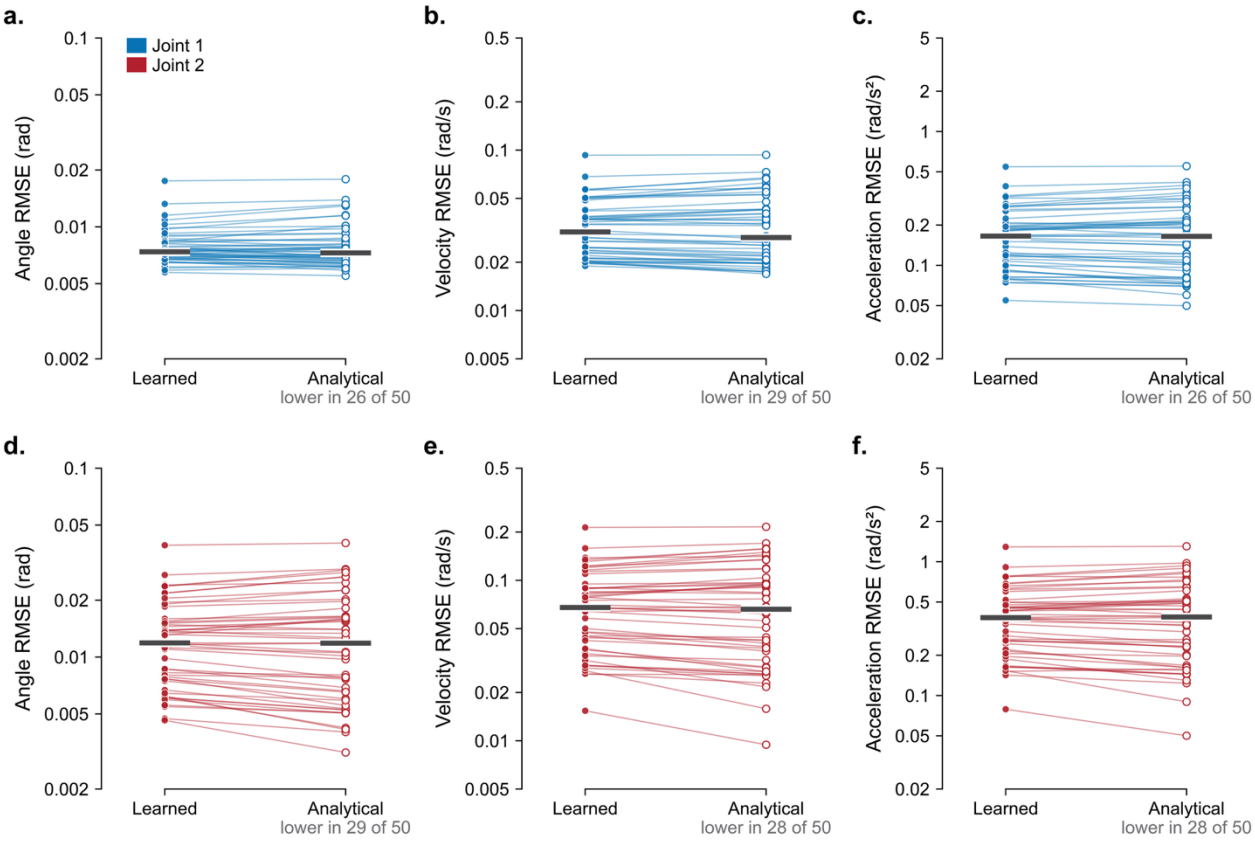
Supplementary Figures and Legends



Supplementary Fig. 1 | Parallel-coordinate representation of hyperparameter optimization. Each polyline represents one Optuna hyperparameter trial evaluated during the architecture search. Axes show the learning rate, activation function, network depth, hidden-layer widths, and validation torque RMSE. Pale cyan lines show all evaluated trials (n = 500), teal lines indicate the lowest 10% of trials by RMSE (n = 50), and the red line marks the best-performing configuration used for subsequent analyses. Unused deeper-layer width parameters in shallower networks are encoded as 0 to display variable-depth architectures on a common axis set.



Supplementary Fig. 2 | Pairwise statistical comparison of inverse-dynamics accuracy across training-set sizes. **a,b**, Matrix representation of pairwise comparisons of trajectory-wise torque RMSE distributions (n = 50 test trajectories per training-set size) for joint 1 (**a**) and joint 2 (**b**) across training-set sizes. Rows and columns denote the dataset sizes being compared. Pairwise differences were evaluated using two-sided Mann–Whitney U tests with Benjamini–Hochberg false discovery rate correction. Cell color reports the FDR-adjusted p value, with darker blue for joint 1 and darker red for joint 2 indicating stronger evidence that the corresponding RMSE distributions differ. Gray diagonal cells represent self-comparisons. Outlined cells labeled ns denote comparisons that were not significant after FDR correction ($p \geq 0.05$).



Supplementary Fig. 3 | Learned and analytical acceleration energies give median closed-loop simulation errors within 8% of each other.

a–f, Root-mean-square error (RMSE) of joint angle (**a**, **d**), velocity (**b**, **e**) and acceleration (**c**, **f**) of the closed-loop forward simulation for joint 1 (**a–c**, blue) and joint 2 (**d–f**, red), on logarithmic axes. Filled circles, solver with the learned acceleration energy (Learned; the errors of Fig. 6b); open circles, the same solver with the analytical Appell acceleration energy substituted for the learned energy (Analytical). Each line joins the two errors of one trajectory (n = 50), heavy bars mark medians, and the count beneath each panel gives the trajectories in which the analytical energy had the lower error. Both simulations used the same solver configuration (16 candidate accelerations within ±15 rad/s^2, 10 Adam refinement steps, λ = 0.1), semi-implicit Euler steps of 2 ms, initial states and fourth-order Runge–Kutta reference trajectories of 5 s. The medians with the analytical energy were 0.92 to 1.01 times those with the learned energy (Kruskal–Wallis tests, p^* = 0.94 for the six comparisons, where p^* is the Benjamini–Hochberg-adjusted p; two-sided Wilcoxon signed-rank tests paired by trajectory, p between 0.41 and 0.98).